

Software Testing & Quality Assurance

Lab Report: Test Case Design using EP and BVA

Project Title: Improved Data Leakage Detection System
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Technology Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)

1. System Under Test: Improved Data Leakage Detection System

The Improved Data Leakage Detection System is a secure enterprise file-sharing platform designed to provide traceability and accountability. The system allows administrators to upload confidential files, generates unique recipient-specific copies using steganographic tracking tokens, and enables forensic leak analysis to attribute compromised files to their original recipients.

2. Requirement FR-1 — Leaked File Forensic Upload Format Validation

During leak investigation, the forensic analysis module extracts steganographic tracking markers from suspected leaked files. The system accepts only supported file types (.txt, .jpg, and .bmp) for extraction. Unsupported formats such as .zip, .mp4, and .docx must be rejected prior to analysis.

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-EP-07
Test Scenario	Verify that the Forensic Leak Analysis module accepts valid suspected file formats and rejects unsupported archive formats like .zip (FR-5).
Preconditions	1. Administrator is logged in and navigates to the Forensic Analysis Module (/admin/investigate).
Test Steps	1. Click on "Upload Leaked File". 2. Select <code>leaked_archive.zip</code> . 3. Click "Extract Forensic Markers".
Input Data	- Leaked File: <code>leaked_archive.zip</code> - File Type: <code>application/zip</code> - Equivalence Partition: P4 (Invalid Compressed Archive)
Expected Result	System forensic parser rejects the upload and displays an error message: <i>"Forensic analysis failed: Unsupported file type. Only original carrier formats (.txt, .jpg, .bmp) can be scanned for tracking tokens."</i> The file is not processed for forensic extraction.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates invalid equivalence partition P4 (Compressed archive) using Equivalence Partitioning (EP) technique.

3. Requirement FR-2 — Forensic Attribution Similarity Match Threshold

When comparing extracted decoy data or steganographic tokens from a recovered file against the database audit logs, the system calculates a match confidence percentage from 0% to 100%. A recipient attribution is confirmed positive only if the match confidence is $\geq 80.0\%$. Matches below 80.0% are marked “Inconclusive / Tampered”.

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-BVA-08
Test Scenario	Verify that forensic leak attribution triggers a positive agent match at exactly 80.0% confidence and classifies scores below 80.0% as inconclusive (FR-5 / BVA).
Preconditions	1. A leaked document <code>confidential_leak.jpg</code> has been uploaded to the forensic engine. 2. Extracted steganographic bits match Agent Hojaifa’s token at exactly 80.0% due to slight compression.
Test Steps	1. Initiate forensic scan for <code>confidential_leak.jpg</code>. 2. Backend calculates token Hamming distance and determines match confidence. 3. Observe the forensic investigation report generated on the dashboard.
Input Data	• Leaked File: <code>confidential_leak.jpg</code> • Extracted Signature Match: 80.0% • Boundary Threshold: 80.0%
Expected Result	System validates that $80.0\% \geq 80.0\%$, confirms positive attribution, and displays: “ <i>Match Confirmed: Recipient copy belongs to Hojaifa (ID: 24524203181) with 80.0% confidence.</i> ”
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates the exact boundary threshold value of 80.0% using Boundary Value Analysis (BVA) technique.

4. Requirement FR-3 — Forensic Bit Error Rate (BER) Tolerance Threshold

When extracting watermarks from a recovered leaked file suffering noise or lossy compression, the Reed-Solomon decoder tolerates a maximum Bit Error Rate (BER) of 5.0%. If BER exceeds 5.0%, the watermark packet is declared unrecoverable.

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-BVA-09
Test Scenario	Verify that the forensic extraction engine blocks recovery when the Bit Error Rate reaches 5.1% (FR-5 / BVA).
Preconditions	1. Recovered leaked file <code>leaked_distorted.jpg</code> has noise resulting in a BER of 5.1%. 2. Administrator initiates forensic token extraction.
Test Steps	1. Navigate to <code>/admin/investigate</code>. 2. Upload <code>leaked_distorted.jpg</code>. 3. Click “Extract Watermark Token”.
Input Data	• Leaked File: <code>leaked_distorted.jpg</code> • Simulated Noise BER: 5.1% • Maximum Recoverable BER: 5.0%
Expected Result	Error correction algorithm halts and returns: <i>“Extraction failed: Bit Error Rate (5.1%) exceeds maximum recoverable threshold of 5.0%. Watermark is damaged.”</i>
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates upper boundary $5.0\% + 0.1\% = 5.1\%$ (over-limit threshold) using Boundary Value Analysis (BVA) technique.